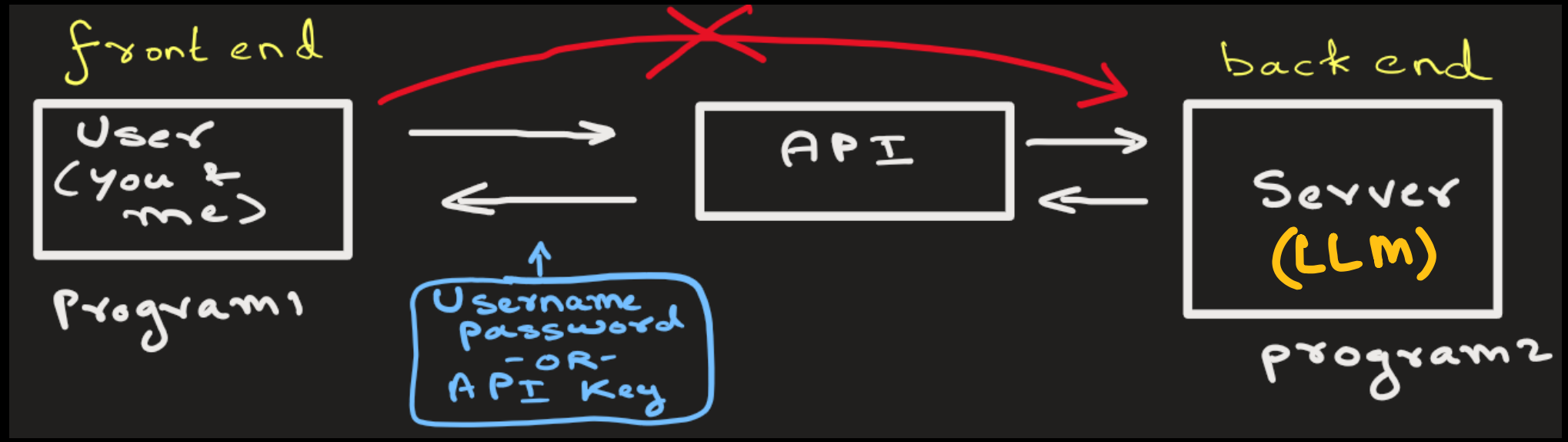
API
(Application Programming Interface)
In LLM

What is an API ?
Why do we need API ?
Examples of API
How to get API Key ?

What is an API ?

API is ,like a middle man, that allows one program application to talk to another program.

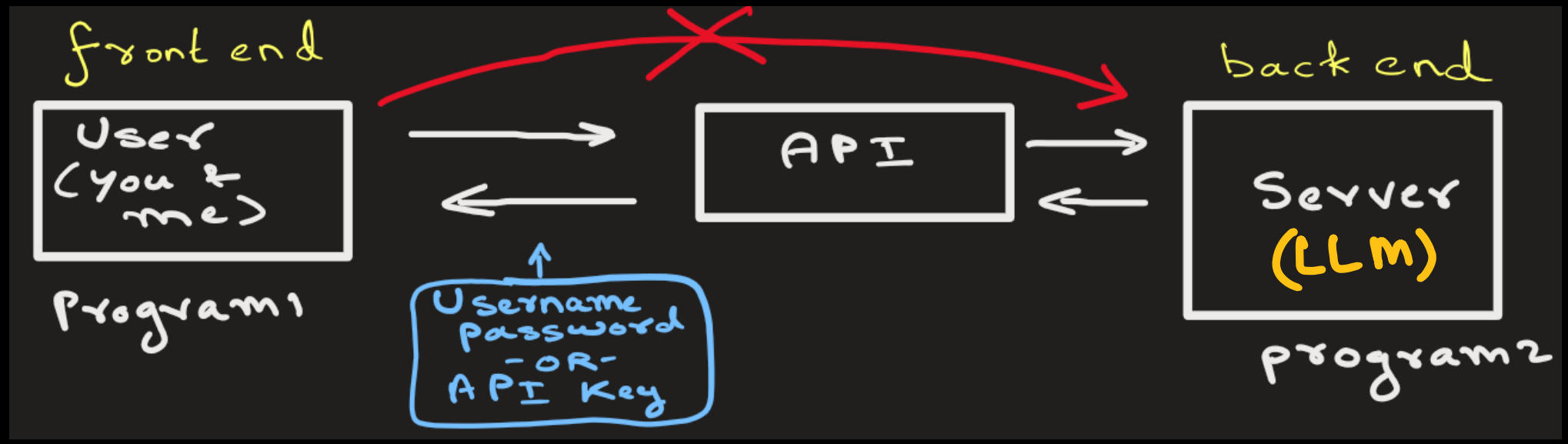
Companies provide API that allows regular user like you and me to access and use their Large Language Models (LLM).
We do not directly talk to their LLM, we use the API.



Generally, to use API you need

- Username and password
- API key

These are provided by the company that is running LLM



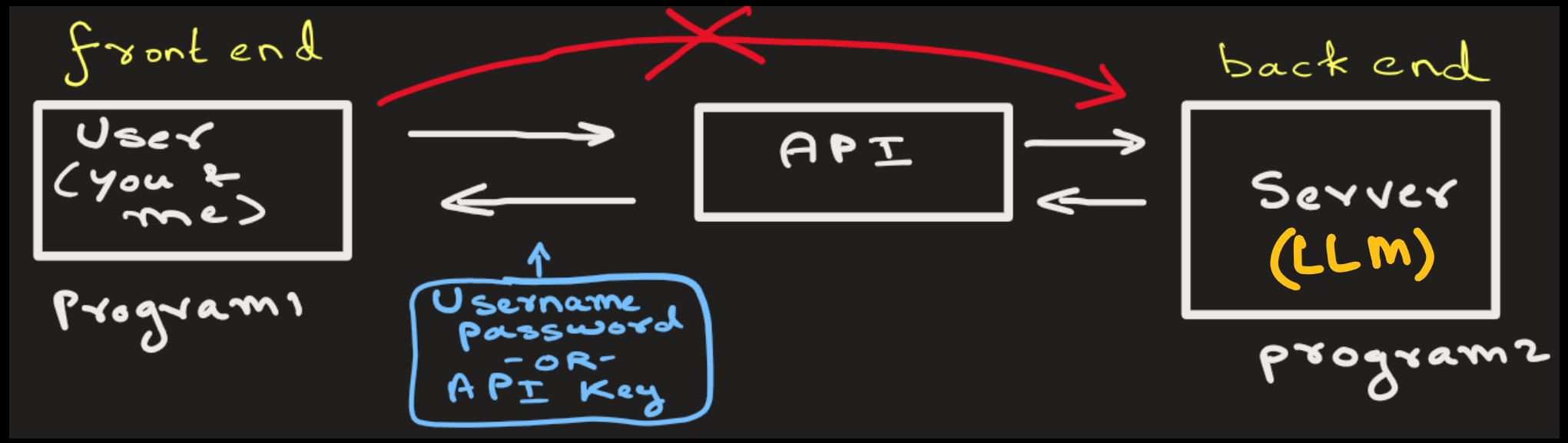
Why do we need an LLM API?

1. Building LLMs is extremely hard

Training something like GPT-4 requires: **Massive data, GPUs (=money), Research expertise**

👉 Not practical for most developers

With an API, developers have an instant access to LLM

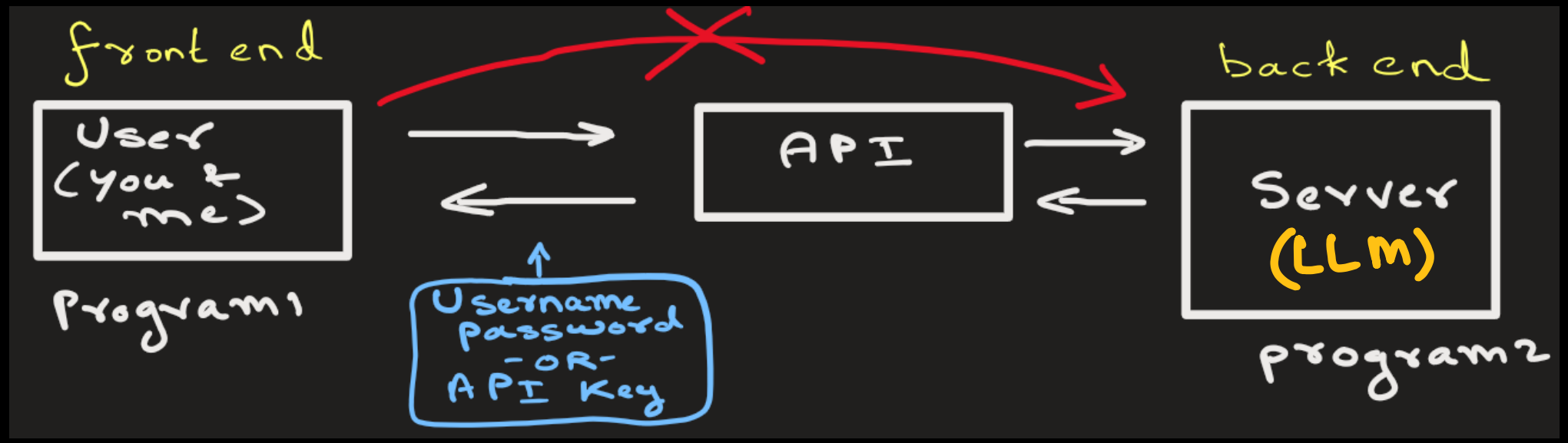


Why do we need an LLM API?

2. Pay-as-you-use

You only pay per request (tokens)

No need to maintain servers

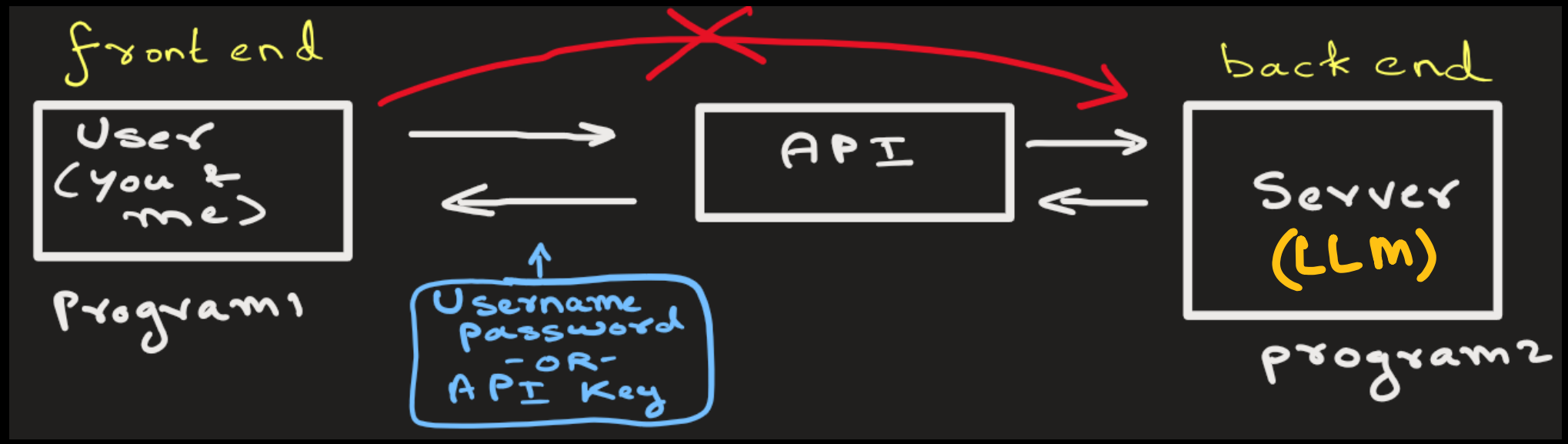


Why do we need an LLM API?

3. API Uses Updated Model

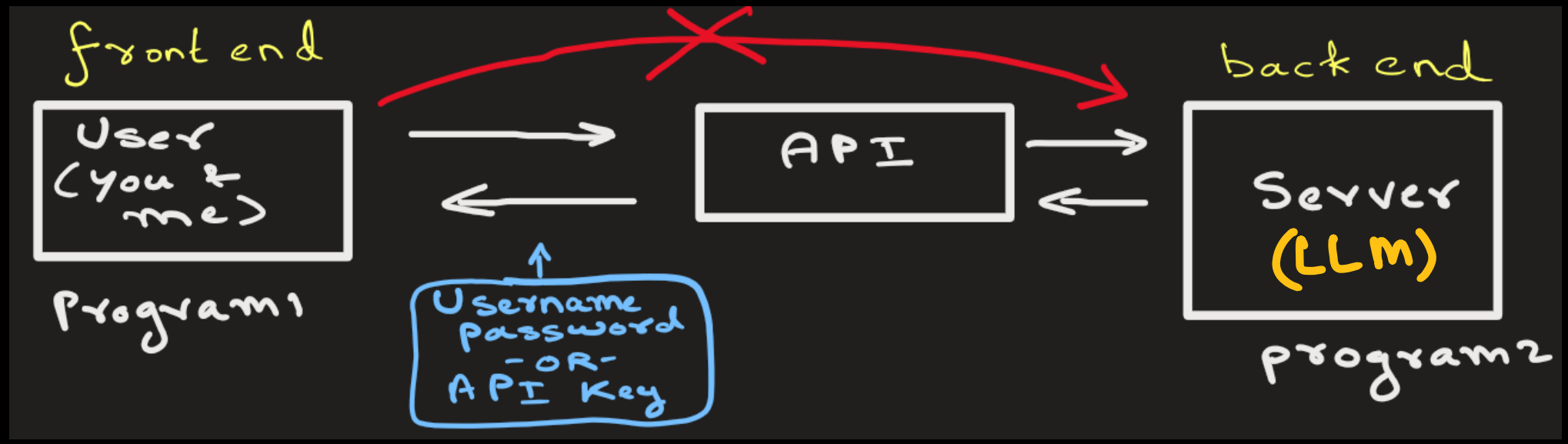
Providers like OpenAI improve models regularly

👉 Your app automatically gets better



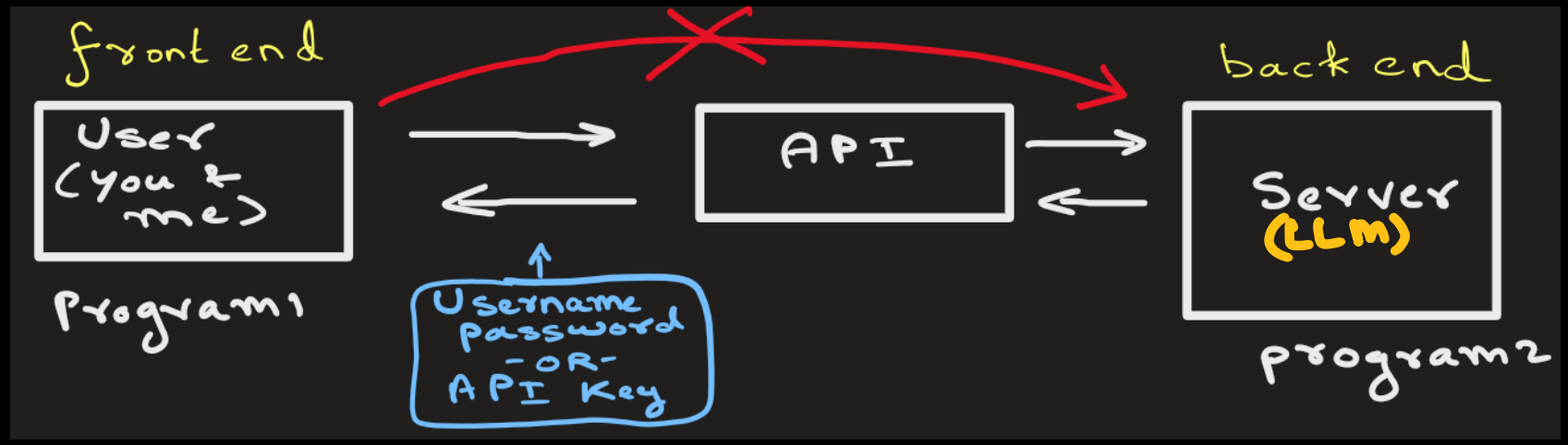
How to use API ?

Ans: Read API –Docs provided by the company.



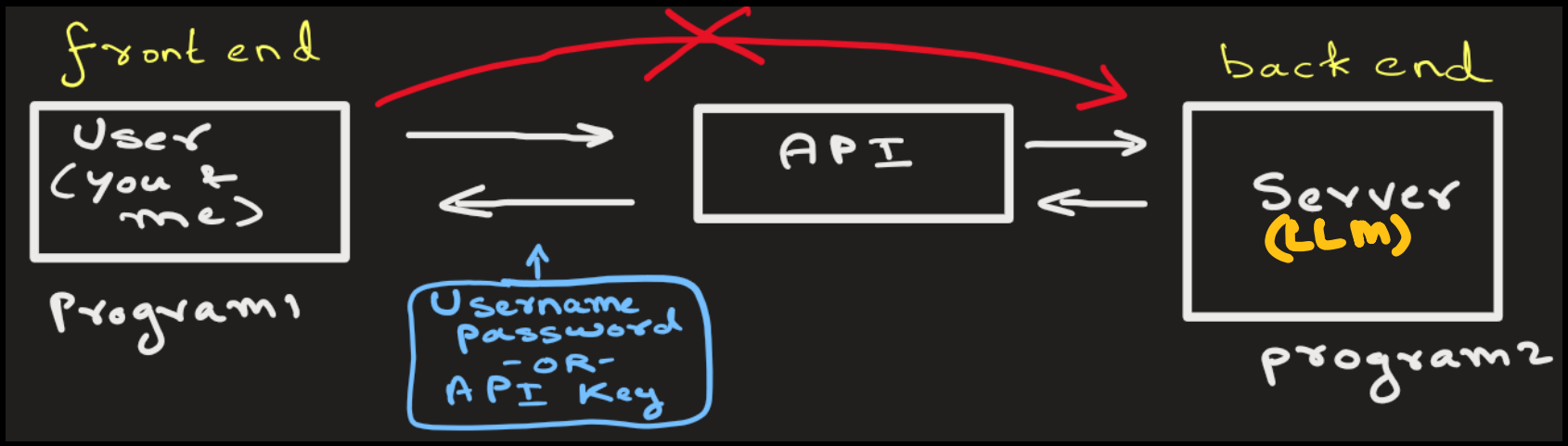
An example:

```python
from openai import OpenAI

# Initialize client
API_KEY="key_here"
client = OpenAI(api_key=API_KEY)

response = client.chat.completions.create(
    model="gpt-4o-mini",
    messages=[
        {"role": "user", "content": "Explain me role of API in LLM in less than 50 words."}
    ],
    temperature=0.7
)

print(response.choices[0].message.content)
```



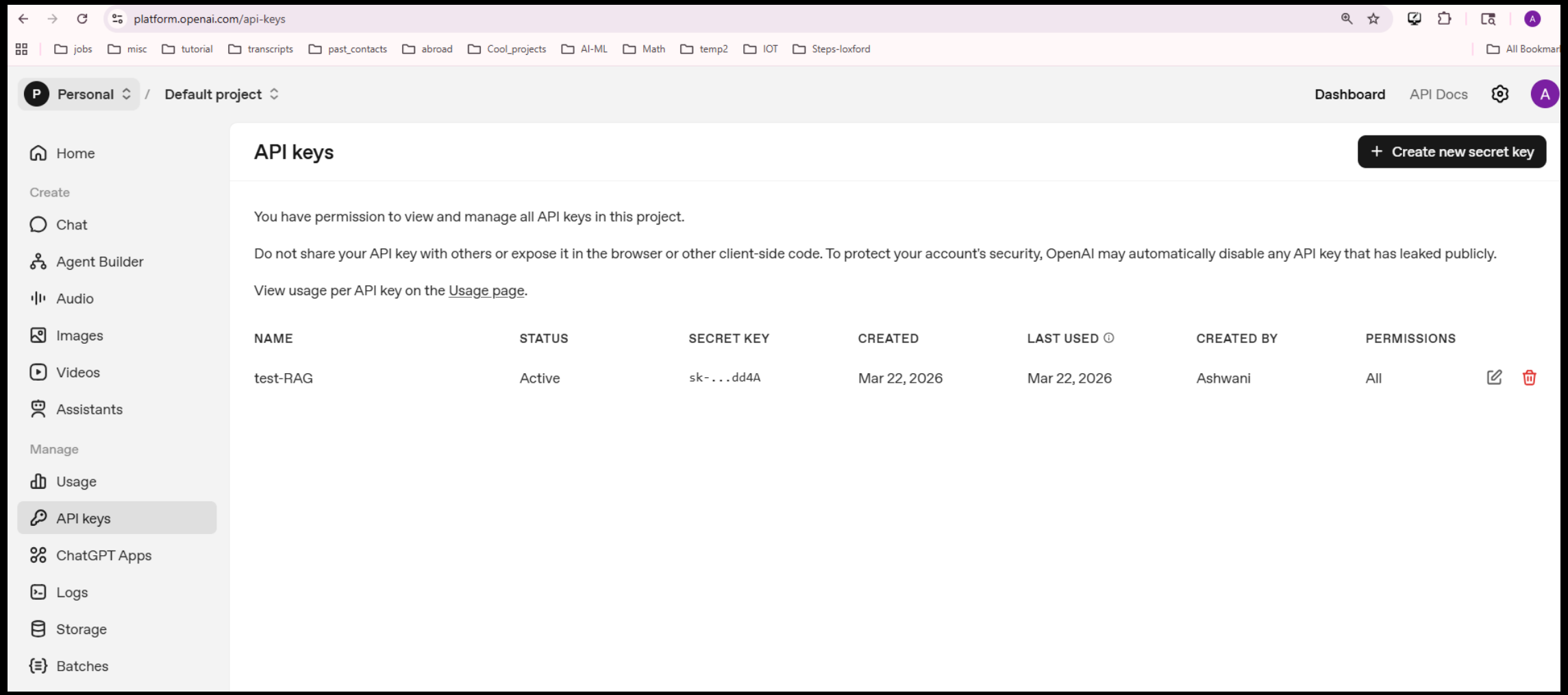
Output:

An API acts as a bridge, allowing external applications to send requests (prompts) to an LLM and receive generated response. It enables developers to integrate powerful AI capabilities into their own software without needing to host or manage the massive LLM themselves.



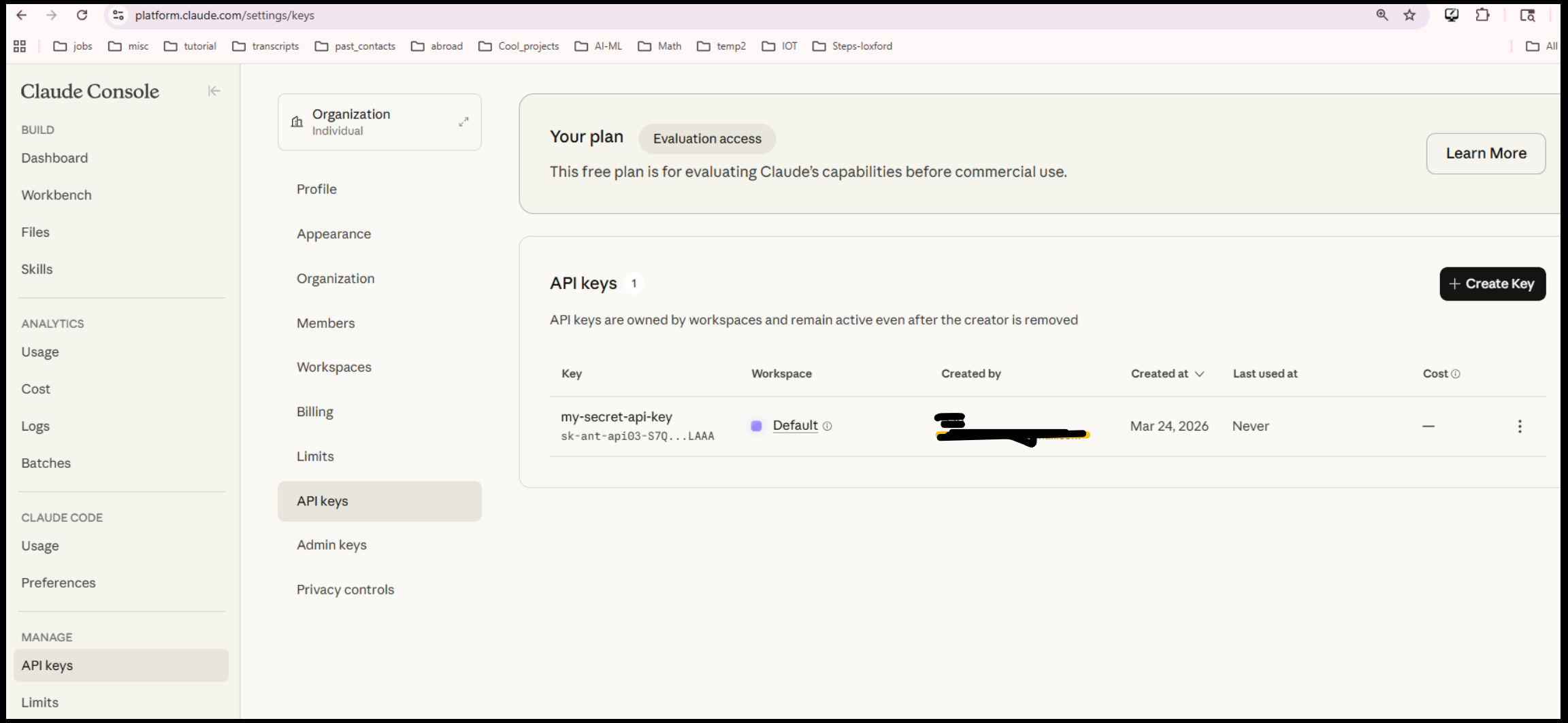
How to get API Key ?

OpenAI API keys : https://platform.openai.com/api-keys



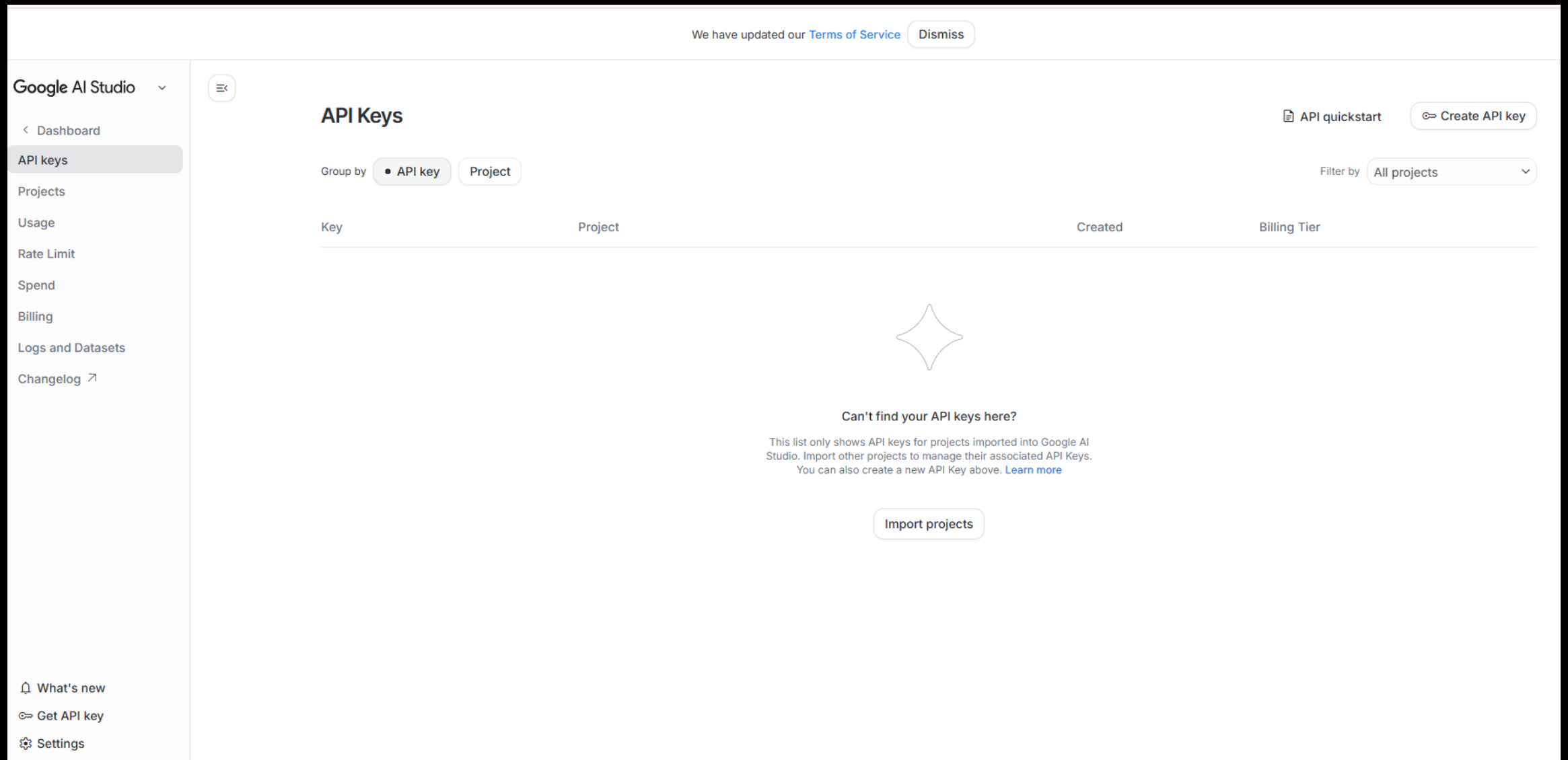
How to get API Key ?

Claude AI API keys : https://platform.claude.com/settings/keys



How to get API Key ?

Google Gemini AI API keys :



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